Viral Genes and Antigens

The genome of HIV contains the **three structural genes** (gag, pol and env) as well as other nonstructural and regulatory genes specific for the virus. These products of these genes, both structural and non structural act as antigens.

Genes coding for structural proteins

1. The gag gene ⟶ Determines the core and shell of the Virus. Precursor protein, p^{55} and it is cleaved into three proteins p^{15}, p^{18} and p^{24}. **Major core antigen p^{24}** can be detected in serum
2. The env gene ⟶ Determines the synthesis of envelope glycoprotein gp^{160}. Cleaved in to gp^{120} and gp^{41}

   The pol gene ⟶ Codes for the **polymerase reverse transcriptase** and other viral enzymes such as protease and endonucleases. It's expressed as a precursor protein, which is cleaved into protein p31, p51 and p66.

Pathogenesis

Infection is transmitted when the Virus **enters the blood or tissues** of a person and comes into contact with a suitable host cell, principally the CD4 lymphocyte. The receptor for the virus is the CD4 antigen and therefore the virus may infect any cell bearing the CD4 antigen on the surface. Specific binding of the **virus to CD4 receptor** is by the envelope **glycoprotein gp^{120}**. Cell fusion is brought about by transmembrane gp^{41}. After **fusion virus** with the host cell membrane, the HIV genome is uncoated and internalized into the cell. Viral reverse transcriptase mediate transcription of its RNA into double stranded DNA, which is **integrated into the genome** of the infected cell through the action of the viral enzyme integrase, causing a latent infection. The primary pathogenic mechanism in HIV infection is the damage caused to the CD4+T lymphocyte. The **T_4 cells decrease** is numbers. Infected T_4 cells do not release normal amounts of interleukin^{-2}, gamma interferon and other lymphokines, this is damping effect on cell mediated immune response.

Clinical Features

AIDS is only the last stage in the wide spectrum in HIV infection.

1. Acute HIV infection

3–6 weeks of infection, persons experience low grade fever, malaise, headache, lymphadenopathy, with rash. Antibodies are usually negative at the onset of the illness but become positive during its course called **'Sero conversion illness'**.

2. Asymptomatic or latent infection

All HIV infected persons, whether or not they experience Sero conversion illness, pass through a phase of **symptomless infection** which may last up to several years. The infection progresses in course of time through various stages, CD4 lymphocytopenia, minor opportunistic infections, ARC AIDS-related complex, ultimately terminating to AIDS.

3. Persistent generalized lymphadenopathy (PGL)

It's defined as the presence of **enlarged lymph nodes** at least 1cm, in diameter in two or more non contiguous extrainguina, sites persist for at least three months.

TN_GOVT_XII_Micro_Biology_CH10.indd 222 04-03-2019 12:27:11